Lightning Impulse Testing

Theory, the h_da impulse voltage generator, the device under test and the measurement

EEN442 Erasmus lab week · High Voltage Laboratory, Hochschule Darmstadt · Wednesday 21 October 2026

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1. Purpose

This handout supports the lightning impulse part of the High Voltage lab. It explains why insulation is tested with lightning impulses, how the standard 1.2/50 µs impulse is defined, how a multi-stage (Marx) impulse generator produces it, and how the impulse is measured. It then describes the generator, the device under test and a recorded measurement from the h_da laboratory.

After the lab you should be able to:

- define the front time T1 and the time to half value T2 and read them from a recorded impulse;
- explain how the stage capacitors are charged in parallel and discharged in series;
- relate the front and tail resistors to the time constants of the impulse;
- distinguish a full impulse from a chopped impulse caused by a breakdown.

2. Why lightning impulse tests

A lightning stroke to or near an overhead line injects a current surge that travels along the conductors as a voltage wave with a steep front, rising within about a microsecond, and a tail that decays within tens of microseconds. Equipment connected to the line, such as transformers, switchgear, insulators and disconnectors, must withstand these overvoltages without breakdown. Because breakdown at microsecond time scales differs from breakdown at power frequency, insulation is qualified with a separate lightning impulse withstand test using a standardised waveform (Betz, Advanced High Voltage Technology, section 2.2).

A typical test sequence from the lecture notes applies 15 impulses of positive and 15 of negative polarity at the rated withstand level, with at most 2 disruptive discharges allowed in self-restoring insulation. As an illustration of the margin involved, a 123 kV gas-insulated switchgear is tested with a 550 kV lightning impulse, about 5.5 times its peak phase-to-earth operating voltage.

3. The standard lightning impulse

IEC 60060-1 defines the standard lightning impulse as a 1.2/50 µs double-exponential waveform. Because the true start of the impulse is hard to locate on a measured record, the time parameters are defined from points on the front:

- **Front time T1:** T1 = 1.67·T, where T is the time between the 30 % and 90 % points of the peak value.
- **Virtual origin O1:** the point where the straight line through the 30 % and 90 % points crosses zero, 0.3·T1 before the 30 % point.
- **Time to half value T2:** the time from O1 until the voltage has fallen to 50 % of the peak on the tail.

Parameter	Standard value	Tolerance
Front time T1	1.2 µs	±30 % (0.84 to 1.56 µs)
Time to half value T2	50 µs	±20 % (40 to 60 µs)
Peak value Û	test level	±3 %
Overshoot		at most 10 % of the peak

Values as summarised by Kalenderli et al. and the Southampton UHV impulse study.

The waveform is the difference of two exponentials with a long tail time constant and a short front time constant. With 68.2 µs and 0.405 µs the curve below gives exactly T1 = 1.20 µs and T2 = 50.0 µs:

$$u(t) = 1.037 \cdot \hat{U} \cdot \left(e^{-t/68.2\,\mu s} - e^{-t/0.405\,\mu s} \right)$$

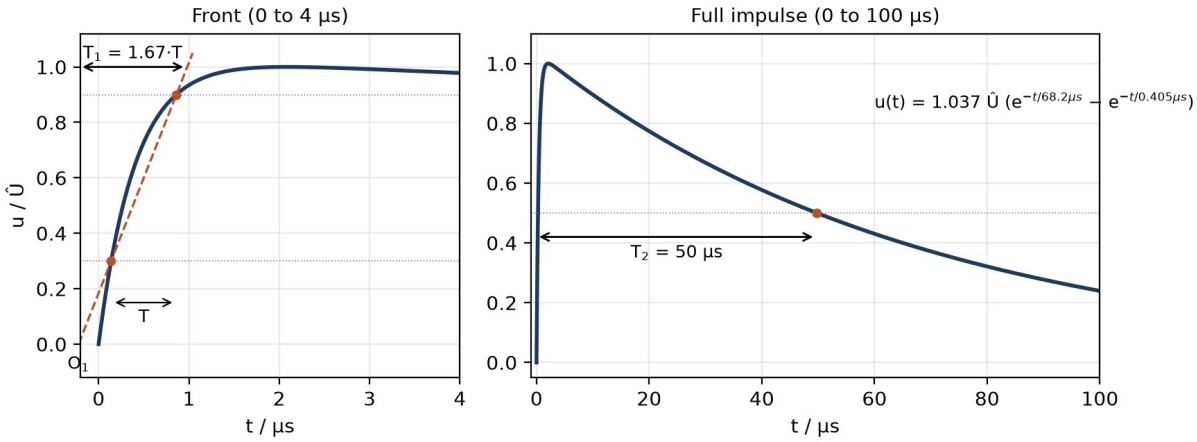


Figure 1. Standard 1.2/50 µs lightning impulse with the T1 and T2 construction of IEC 60060-1 (computed from the expression above).

A chopped impulse is a full impulse that collapses suddenly, for example because of a flashover of the test object or the deliberate firing of a chopping gap. Tail-chopped impulses (chopped a few microseconds after the peak) are especially severe for windings, such as those of instrument transformers, because the sudden collapse produces a very steep voltage change.

4. Single-stage impulse circuits

The basic impulse circuit stores energy in an impulse capacitor CS, charged to U0 through a charging resistor RL. When the spark gap FS fires, CS discharges into the load capacitance CB (test object, divider and stray capacitance) through the damping or front resistor RD, while the discharge or tail resistor RE drains the charge and forms the tail. The notes use two arrangements that differ in the position of RE.

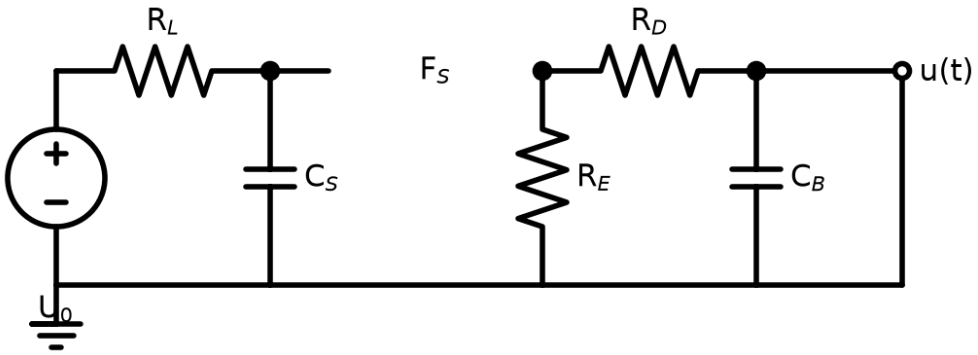


Figure 2. Single-stage impulse circuit, arrangement B (RE directly across CS). In arrangement A, RE is connected across CB after RD.

	Arrangement A (RE after RD)	**Arrangement B (RE before RD)**
Front time constant	(RE·RD/(RE+RD))·(CS·CB/(CS+CB))	RD·CS·CB/(CS+CB)
Tail time constant	(RE+RD)·(CS+CB)	RE·(CS+CB)
Efficiency η = Û/U0	(RE/(RE+RD))·(CS/(CS+CB))	CS/(CS+CB) (higher)

For the usual case RE·CS >> RD·CB the time parameters follow directly from the two time constants:

$$T1 \approx 2.96 \times \text{front time constant} \qquad T2 \approx 0.73 \times \text{tail time constant}$$

For arrangement B this gives T1 ≈ 2.96·RD·CS·CB/(CS+CB) and T2 ≈ 0.73·RE·(CS+CB).

So the front resistor RD sets the front time and the tail resistor RE sets the time to half value. To keep the efficiency high and make the shape insensitive to the test object, the notes recommend CS ≥ 10·CB and CB ≥ 10·C_TO, where C_TO is the test-object capacitance and CB includes the divider, the test object and the connections.

5. The multi-stage Marx generator

A single capacitor and gap cannot practically be charged to several hundred kilovolts. The Marx principle, proposed by Erwin Otto Marx in 1923 (Marx generator), solves this with n stages:

1. Charging: the n stage capacitors CS' are charged in parallel from one DC supply U0 through the charging resistors RL', which take seconds to about a minute.
2. Erection: when the first sphere gap is triggered, the potential of the next stage jumps, its gap is overstressed and fires, and the breakdown runs up the column. All gaps fire within a fraction of a microsecond and the capacitors are now in series.
3. Discharge: the series capacitors, with total charge voltage n·U0, discharge through the front and tail resistors into the load. The charging resistors are large enough to act as an open circuit during the impulse.

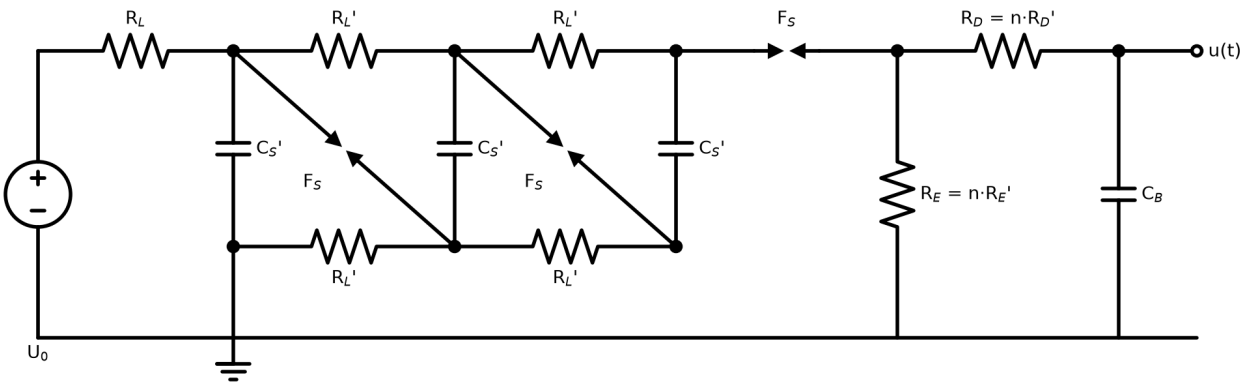


Figure 3. Simplified three-stage Marx generator. In real generators the front and tail resistors are usually distributed over the stages.

The equivalent single-stage values of an n-stage generator with stage values CS', RD', RE' are:

$$CS = CS'/n \qquad RD = n \cdot RD' \qquad RE = n \cdot RE' \qquad \hat{U} \approx \eta \cdot n \cdot U0$$

The stored energy $n \cdot ½ \cdot CS' \cdot U0^2$ is the usual rating of an impulse generator.

Illustrative design example

Target: a 1.2/50 µs impulse into CB = 1 nF with an n = 4 stage generator, CS' = 40 nF per stage (CS = 10 nF), using arrangement B. These values are illustrative only and are not those of the h_da generator or of the exercise in the lecture notes.

Quantity	Calculation	Result
Front resistor	RD = T1/(2.96·CS·CB/(CS+CB)) = 1.2 µs/(2.96·0.909 nF)	446 Ω (111.5 Ω per stage)
Tail resistor	RE = T2/(0.73·(CS+CB)) = 50 µs/(0.73·11 nF)	6.23 kΩ (1.56 kΩ per stage)
Check	RE·CS = 62 µs >> RD·CB = 0.45 µs	approximation valid
Efficiency	η = CS/(CS+CB)	0.909
Peak at U0 = 100 kV per stage	$\hat{U} = \eta \cdot n \cdot U0$	about 364 kV
Stored energy	$n \cdot ½ \cdot CS' \cdot U0^2$	800 J

6. The h_da impulse voltage generator



Figure 4. Impulse voltage generator in the h_da High Voltage Laboratory.

The lecture notes describe the h_da generator as a unit for impulse voltages up to 400 kV, built from 4 units connected in series (Betz, figure 2.8).

In the photo the copper spheres of the stage gaps are stacked along the vertical axis. During the lab, count the stages and compare them with the four units quoted in the notes.

Components visible in the photo:

- **Sphere gaps:** copper sphere pairs, one per stage, which switch the stages from parallel to series.
- **Gap-spacing drive:** the white insulating rod along the centre, which probably sets the sphere spacing and thus the firing voltage.

- **Stage resistors:** the blue diagonal rods between the levels, probably the charging and wave-shaping resistors.
- **Stage capacitors and supports:** the dark brown ribbed columns in the upper frame and the grey ribbed columns at the base, either stage capacitors or insulating supports.
- **Right-hand column:** the column with copper grading discs, linked by horizontal tubes, possibly a divider or resistor column.
- **Earthing:** wide copper foil strips on the floor provide a low-inductance earth return, which is essential for clean impulse shapes.

7. Device under test



Figure 5. Test set-up: the device under test in front of the impulse generator (right).

The test object carries the label RPS TracFeed. TracFeed is the product line of Rail Power Systems GmbH for railway power supply. The construction in the photos, a moving contact arm on a blue composite insulator with copper contacts, mounted on a base frame, matches the TracFeed MTS mast disconnector, a switch used to connect and disconnect feeder sections of the railway overhead contact line (Rail Power Systems).

According to the manufacturer, the MTS family has variants for 3 kV DC and 15/25 kV AC, uses dirt- and water-repellent composite insulators (creepage distance 1590 mm for the 25 kV models), is tested for at least 50 000 switching cycles and is designed to DIN EN 50152-2 (TracFeed MTS brochure).



Figure 6. Close-up of the disconnector: fixed contact on the grey support (left) and moving contact on the composite insulator (right).

In the photos the disconnector appears to be open: the copper contact on the composite insulator is turned away from the fixed contact on the grey support. A lightning impulse test on a disconnector stresses both the insulation to earth (along the composite insulator) and the isolating distance between the open contacts, which must withstand a higher level so that the open switch remains a safe isolation point. The lab staff will explain which terminal is energised and which test level is applied.

Behind the insulator stands a clear cylindrical column with blue elements and an aluminium toroid on top, probably the impulse voltage divider with its grading ring. The toroid smooths the electric field at the high-voltage top and prevents corona.

8. Measurement

The impulse voltage is reduced by the voltage divider and recorded with an HBK (formerly HBM) Genesis HighSpeed transient recorder controlled by the Perception software. In such systems the digitiser near the divider is connected to the control room by optical fibre, so the measurement is fully isolated from the test hall, and the software evaluates the peak value and time parameters according to IEC 60060-2 (HBK case study, KEMA laboratory).



Figure 7. Recorded impulse in Perception 7.20, 21 October 2026 at 11:28 (recording hda_2025_11_00019).

Settings visible on the screen:

- **Time base:** 1 µs per division, window from −1.6 µs to 8.4 µs.
- **Vertical range:** ±300 kV.
- **Sampling rate:** 100 MS/s, one sample every 10 ns, so about 120 samples lie on a 1.2 µs front.
- **Acquisition:** multi-segment mode with 500 segments of 1 ms; the status "Auf den Trigger 8 von 500 wird gewartet" means the system is waiting for trigger 8 of 500, so each impulse of the series is stored as its own segment.
- **Optical link:** the tab "Status des LWL" (Lichtwellenleiter) shows the status of the fibre-optic connection.

Reading the record:

- **Front:** the voltage rises within roughly a microsecond, consistent with a lightning impulse front. The red sloped line on the front appears to be the 30 % to 90 % construction used to find T1.
- **Crest:** the voltage stays almost constant after the peak because, at 1 µs per division, only the first few microseconds of a 50 µs tail are shown.
- **Collapse:** after about 2 to 3 µs the voltage drops abruptly to zero and rings briefly. This is a chopped impulse. It is caused either by a flashover of the device under test or by the firing of a chopping gap; note which one during the lab.
- **Cursors:** the horizontal cursors read 154.9 kV and 110.9 kV. The operator places the cursors to read voltage levels on the record.

In the lab report, use the recorded impulses to determine Û, T1 and, for full impulses, T2, compare them with the tolerances of section 3, and identify which impulses of the series led to a breakdown.

9. Safety

- Enter the test area only when the generator is switched off, all stages are earthed with the earthing rod and the lab staff has released the area.
- Capacitors can recharge from dielectric absorption after earthing. Keep the earthing in place while working in the test area.
- Follow the interlocks of the cage and the instructions of Prof. Betz and Mr Radler at all times.

References

- T. Betz, Advanced High Voltage Technology, lecture notes, h_da, section 2.2: https://sps-l.github.io/hda-labweek-2026/files/HV_Lecture_Notes_Betz.pdf
- O. Kalenderli et al., lightning impulse parameter evaluation, EMC Europe 2002: https://web.itu.edu.tr/kalenderli/publ/paper/2002-EMC_Europe02_Sorrento_Italy-AA-OKalenderli-AIY-HK.pdf
- Southampton, UHV lightning impulse paper: https://eprints.soton.ac.uk/267056/1/UHV_paper_Long.pdf
- Transformers Magazine, insulation design e-lesson 9: https://transformers-magazine.com/files/Design-of-power-transformers-e-lesson-9-Insulation-design-SLIDES-FOR-THE-ATTENDEES.pdf
- Marx generator, Wikipedia: https://en.wikipedia.org/wiki/Marx_generator
- Impulse generator, Wikipedia: https://en.wikipedia.org/wiki/Impulse_generator
- Rail Power Systems, TracFeed MTS mast disconnector: https://www.rail-ps.com/en/systems-and-products/procucts/tracfeed-mts-mast-disconnector
- Rail Power Systems, TracFeed MTS brochure 2026: https://www.rail-ps.com/content/broschueren/437_EN_TracFeed%C2%AE_MTS_2026.pdf
- HBK, KEMA High Voltage Laboratory case study (Genesis HighSpeed and Perception): https://www.hbkworld.com/fr/knowledge/resource-center/case-studies/kema-high-voltage-lab-using-hbms-genesis-high-speed

Standards: IEC 60060-1 High-voltage test techniques, Part 1: General definitions and test requirements; IEC 60060-2 Part 2: Measuring systems; DIN EN 50152-2 Railway applications, fixed installations, particular requirements for AC switchgear, Part 2: disconnectors, earthing switches and switches.